



Maulana Azad National Institute of Technology Bhopal

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Assignment 3 EM fields and Materials Vidhi Sharma 2311301120

1)

a) Given, $k_1 = 2.5$
 $k_2 = 1$

$$\vec{D}_1 = 12\hat{x} - 10\hat{y} + 4\hat{z} \text{ nC/m}^2$$

x - component is normal component
 $\Rightarrow D_{1n} - D_{2n} = \rho_s$

as there is no charge at interface,
 $D_{1n} = D_{2n}$

$$\vec{D}_2 = \cancel{12\hat{x}} 12\hat{x} + D_{2y}\hat{y} + D_{2z}\hat{z} \quad (i)$$

and y and z are tangential components,

$$\vec{D}_2 = \epsilon_1 E_1 = k_2 \epsilon_0 E_1$$

$$E_1 = \frac{\vec{D}_1}{k_1 \epsilon_0} = \frac{12\hat{x}}{2.5\epsilon_0} - \frac{10\hat{y}}{2.5\epsilon_0} + \frac{4\hat{z}}{2.5\epsilon_0}$$

According to boundary condition,
 $E_{1t} = E_{2t}$

$$\vec{E}_2 = E_{2x}\hat{x} - \frac{10}{2.5\epsilon_0}\hat{y} + \frac{4}{2.5\epsilon_0}\hat{z}$$

$$\vec{D}_2 = \epsilon_0 E_2 = \epsilon_0 E_2 x \hat{x} - 4 \hat{y} + \frac{4}{2.5} \hat{z} \quad (ii)$$

From (i) & (ii),

$$12 \hat{x} + D_{2y} \hat{y} + D_{2z} \hat{z} = \epsilon_0 E_2 x \hat{x} - 4 \hat{y} + \frac{4}{2.5} \hat{z}$$

$$\vec{D}_2 = 12 \hat{x} - 4 \hat{y} + 1.6 \hat{z} \quad \text{nC/m}^2$$

$$\vec{D}_2 \cdot \hat{a}_x = |\vec{D}_2| (1) \cos \theta_2$$

$$12 = 12.75 \cos \theta_2$$

$$\cos \theta_2 = 0.941 \Rightarrow \theta_2 = 19.75^\circ$$

b) Given, $E_2 = 12 \text{ V/m}$, $\theta_2 = 60^\circ$

$$E_{2t} = E_2 \sin \theta_2 = E_{1t} = E_1 \sin \theta_1$$

$$E_{1t} = 12 \sin 60^\circ = 6\sqrt{3} \text{ V/m}$$

$$D_{1n} = D_{2n} = \epsilon_0 E_{2n} = k_1 \epsilon_0 E_{1n}$$

$$\frac{\tan \theta_1}{\tan \theta_2} = \frac{\epsilon_{x1}}{\epsilon_{x2}} \Rightarrow \tan \theta_1 = \frac{2.5}{1} \times \sqrt{3}$$

$$\theta_1 = 77^\circ$$

$$E_1 = \frac{E_{1t}}{\sin \theta_1} = 10.7 \text{ V/m}$$



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2)

a) Point A is in the conductor,
 $\Rightarrow E = D = 0$

b) Point B $(-4, 2, 5)$ is in dielectric medium,
 $\Rightarrow D_n = \rho_s = 2 \text{ nC/m}^2$

$$\vec{D} = 2\hat{y} \text{ nC/m}^2$$

$$\vec{E} = \frac{\vec{D}}{\epsilon_0 \epsilon_r} = \frac{2 \times 36\pi \times 10^{-9} \times 10^9 \hat{y}}{2}$$

$$\vec{E} = 113.1\hat{y} \text{ V/m}$$

3)

a) $V = \frac{4yz}{x^2+1}$ at $P(2, 2, 3)$

$$|V_P| = \frac{4 \times 2 \times 3}{1+1} = 12 \text{ V}$$

$$\vec{E} = -\nabla V$$

$$= -\left[-4yz \times \frac{2x}{(1+x^2)^2} \hat{a}_x + \frac{4z}{x^2+1} \hat{a}_y + \frac{4y}{x^2+1} \hat{a}_z \right]$$

$$\rho_v = \nabla \cdot \vec{D} = \nabla \cdot (\epsilon_0 \vec{E})$$

$$\rho_v = 4yz \left(\frac{2-4x^2-6x^4}{(1+x^2)^2} \right) - 0 - 0$$

$$\rho_v = \frac{4 \times 2 \times 3 (2-4-6)}{4} = -48 \text{ C/m}^3$$

b) $V = 5\rho^2 \cos 2\phi$ at $P(\rho=3, \phi=\frac{\pi}{3}, z=2)$

$$|V| = \frac{-45}{2} = -22.5 \text{ V}$$

$$\epsilon (\nabla^2 V) = -\rho_v$$

$$\frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2} = -\frac{\rho_v}{\epsilon}$$

$$20 \cos 2\phi + (-20 \cos 2\phi) + 0 = -\frac{\rho_v}{\epsilon}$$

$$\rho_v = 0$$

c) $V = \frac{2 \cos \phi}{r^2}$ at $P(r=0.5, \theta=45^\circ, \phi=60^\circ)$

$$|V| = \frac{2 \cos 60^\circ}{(0.5)^2} = 4 \text{ V}$$

$$\nabla^2 V = -\rho_v / \epsilon$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = -\frac{\rho_v}{\epsilon}$$

$$\frac{4 \cos \phi}{r^4} + 0 + \left(\frac{-2 \cos \phi}{r^4 \sin^2 \theta} \right) = -\frac{\rho_v}{\epsilon}$$

$$\frac{2}{0.5^4} - \frac{2}{0.5^4} = 0 \Rightarrow \rho_v = 0$$



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4. Inner conductor $\Rightarrow V_1 = 0, \epsilon_1$
Outer conductor $\Rightarrow V_2 = V_0, \epsilon_2$
 $\rho_v = 0$

From Laplace equation,

$$\nabla^2 V = 0$$

$$[c < r < a]$$

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right)$$

$$+ \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2} = 0$$

$$\Rightarrow \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) = 0$$

Integrating,

$$r^2 \frac{\partial V}{\partial r} = A_1$$

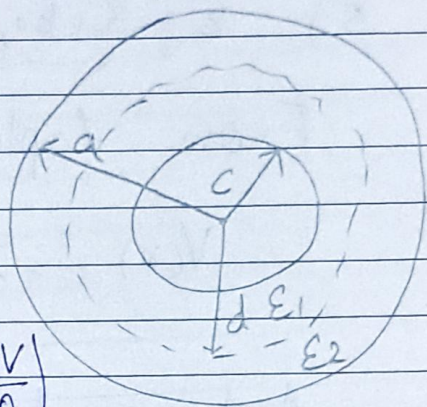
$$V(r) = \frac{A}{r} + B$$

Applying boundary conditions,

$$V(c) = 0 = \frac{A}{c} + B \Rightarrow A = -Bc$$

$$V(a) = V_0 = \frac{A}{a} + B \Rightarrow A = (V_0 - B)a$$

$$\Rightarrow B = \frac{aV_0}{a-c} \quad \& \quad A = -\frac{acV_0}{a-c}$$



For $c < r < d$,

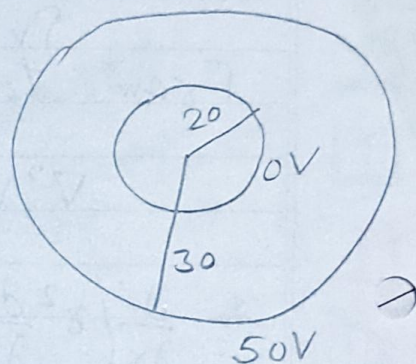
$$\Rightarrow V(r) = -\frac{acV_0}{(a-c)r} + \frac{aV_0}{a-c}$$

For $d < r < a \Rightarrow V(r) = V_0(\epsilon_1/\epsilon_2) \left(\frac{a}{a-c} - \frac{ac}{a-c} \right)$

5) $\epsilon = 3.1 \epsilon_0$, $\sigma = 10^{-12} \text{ S/m}$
 $\rightarrow$ conductivity

From Laplace equation,

$$V(r) = \frac{A}{r} + B$$



Applying boundary conditions,

$$V(20) = \frac{A}{20} + B = 0 \Rightarrow A = -20B$$

$$V(30) = \frac{A}{30} + B = 50 \Rightarrow A = (50 - B)30$$

$$\Rightarrow A = -3000 \text{ Vcm}, \quad B = 150 \text{ V}$$

$$\Rightarrow V(r) = 150 - \frac{3000}{r} = 150 \left(1 - \frac{20}{r} \right) \text{ V}$$

$$E(r) = -\frac{dV}{dr} = -\frac{3000}{r^2} \text{ V/m}$$

$$D(r) = \epsilon E(r) = 3.1 \epsilon_0 \left(\frac{-3000}{r^2} \right) \text{ C/m}^2$$



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b) Charge densities,

$$D(20) = 3.1 \epsilon_0 \left(\frac{-3000}{20^2} \right) \times 10^4 \text{ C/m}^2$$
$$= -2.05 \text{ } \mu\text{C/m}^2$$

$$D(30) = -3.1 \epsilon_0 \left(\frac{-3000}{30^2} \right) \times 10^4 \text{ C/m}^2$$
$$= 0.91 \text{ } \mu\text{C/m}^2$$

c) Leakage resistance,

$$R = \frac{\rho}{4\pi} \left(\frac{1}{r_1} - \frac{1}{r_2} \right)$$

$$\rho = \frac{1}{\sigma}$$

$$R = \frac{10^{12}}{4\pi} \left(\frac{10^{+2}}{20} - \frac{10^{+2}}{30} \right) \Omega$$

$$R = 1.3 \times 10^{11} \Omega$$